

Circles

EE25BTECH11008 - Anirudh M Abhilash

October 11, 2025

Problem Statement

The lines $2x - 3y = 5$ and $3x - 4y = 7$ are diameters of a circle having area 154 sq.units. Find the equation of the circle.

Solution

The equations of the diameters are

$$2x - 3y = 5, \quad (1)$$

$$3x - 4y = 7. \quad (2)$$

$$\begin{pmatrix} 2 & -3 \\ 3 & -4 \end{pmatrix} \mathbf{c} = \begin{pmatrix} 5 \\ 7 \end{pmatrix}, \quad (3)$$

Substituting $\mathbf{u} = -\mathbf{c}$:

$$\begin{pmatrix} 2 & -3 \\ 3 & -4 \end{pmatrix} \mathbf{u} = -\begin{pmatrix} 5 \\ 7 \end{pmatrix}. \quad (4)$$

Solution (cont..)

Augmented matrix of the system:

$$\left(\begin{array}{cc|c} 2 & -3 & -5 \\ 3 & -4 & -7 \end{array} \right).$$

Step 1:

$$R_2 \rightarrow 3R_1 - 2R_2$$

$$\left(\begin{array}{cc|c} 2 & -3 & -5 \\ 0 & -1 & -1 \end{array} \right).$$

Step 2:

$$R_2 \rightarrow -R_2$$

$$\left(\begin{array}{cc|c} 2 & -3 & -5 \\ 0 & 1 & 1 \end{array} \right).$$

Solution (cont..)

Step 3:

$$R_1 \rightarrow R_1 + 3R_2$$
$$\left(\begin{array}{cc|c} 2 & 0 & -2 \\ 0 & 1 & 1 \end{array} \right).$$

Step 4:

$$R_1 \rightarrow \frac{1}{2}R_1$$
$$\left(\begin{array}{cc|c} 1 & 0 & -1 \\ 0 & 1 & 1 \end{array} \right).$$

Thus the solution is

$$\mathbf{u} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}. \quad (5)$$

Solution (cont..)

From the area,

$$\pi r^2 = 154 \quad \Rightarrow \quad r^2 = 49 \quad \Rightarrow \quad r = 7. \quad (6)$$

Compute f (using $f = \|\mathbf{u}\|^2 - r^2$):

$$\|\mathbf{u}\|^2 = (-1)^2 + 1^2 = 2, \quad (7)$$

$$f = 2 - 49 = -47. \quad (8)$$

The general quadratic form is

$$\mathbf{x}^T V \mathbf{x} + 2\mathbf{u}^T \mathbf{x} + f = 0, \quad (9)$$

and for a circle $V = I_{2 \times 2}$. Substituting the values:

$$\mathbf{x}^T I \mathbf{x} + 2 \begin{pmatrix} -1 & 1 \end{pmatrix} \mathbf{x} - 47 = 0. \quad (10)$$

Solution (cont..)

Expanded,

$$x^2 + y^2 - 2x + 2y = 47, \quad (11)$$

$$x^2 + y^2 - 2x + 2y = 47$$

C Code (Computations)

https:

`//github.com/Anirudh-EE25BTECH11008/ee1030-2025/tree/
main/EE25BTECH11008/MATGEO/7.4.15/Codes/circle.c`

Python Code

https:

`//github.com/Anirudh-EE25BTECH11008/ee1030-2025/tree/
main/EE25BTECH11008/MATGEO/7.4.15/Codes/circle.py`

Plot

